

 Reliable INSTITUTE Unleashing Potential	MATHEMATICS REVISION PLAN-3 Target : JEE (Advanced)-2023	DPP DAILY PRACTICE PROBLEM DPP NO. # 06 (Advanced)
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Max. Time : 129 Minutes **Total Marks-166**

Q. 01 to 06	Comprehension	('–1' negative marking)	[3 Marks,3 Min.][18,18]
Q. 07 to 26	Multiple choice objective	('–1' negative marking)	[4 Marks,3 Min.][80,60]
Q. 27 to 28	Match the column	(no negative marking)	[4 Marks,3 min.][8,6]
Q. 29 to 43	Numerical Type	('–1' negative marking)	[4 Marks,3 Min.][60,45]

DPP Syllabus : Binomial Theorem, Permutation & Combination and Probability

Comprehension # 1 (Q.1 to Q.2)

$$\text{Consider the function } f(x, n) = \frac{\sum_{r=0}^{\left[\frac{n}{2}\right]} {}^nC_{2r} x^r}{\sum_{r=0}^{\left[\frac{n-1}{2}\right]} {}^nC_{2r+1} x^r} \quad (\text{where } [.] \text{ denotes greatest integral function}).$$

1. The value of $f(9, 10)$ is

(A) $3 \left(\frac{2^{10} + 1}{2^{10} - 1} \right)$ (B) $\frac{1}{2} \left(\frac{2^{10} + 1}{2^{10} - 1} \right)$ (C) $3 \left(\frac{4^{70} + 1}{2^{10} - 1} \right)$ (D) $\frac{1}{3} \left(\frac{4^{10} + 1}{4^{10} - 1} \right)$

2. If $f(100, 101) = \frac{f(100, 100) + 100}{f(100, 100) + \lambda}$, then the value of λ is

(A) –1 (B) 1 (C) 0 (D) 4

Comprehension # 2 (Q.3 to Q.4)

Let $1, \alpha_1, \alpha_2, \dots, \alpha_k$ are divisors of number $N = 2^{n-1}(2^n - 1)$, where $2^n - 1$ is a prime number and $1 < \alpha_1 < \alpha_2 < \dots < \alpha_k$.

3. If the value of n can be chosen from the set of first ten natural numbers, then the value of

$$\frac{\alpha_1 + \alpha_2 + \dots + \alpha_k}{k} \text{ can be}$$

(A) $\frac{240}{7}$ (B) $\frac{239}{7}$ (C) $\frac{992}{9}$ (D) $\frac{991}{9}$

4. The value of $1 + \frac{1}{\alpha_1} + \frac{1}{\alpha_2} + \dots + \frac{1}{\alpha_k}$ is

(A) 2 (B) 3 (C) k (D) $k + 1$

Comprehension # 3 (Q.5 to Q.6)

In a game, there are four shells. Each shell has a concave side and convex side. In this each player has to throw all four shells together. The player will get one point for each concave shell and zero points for convex shell. Also if all shells are seen convex, then the player will get 8 points.

5. Find the probability of scoring 14 points in 3 throws.

(A) $\frac{5}{216}$ (B) $\frac{11}{512}$ (C) $\frac{11}{256}$ (D) $\frac{21}{1024}$

6. If a score of 12 was obtained in 4 throws, then find the probability of showing three concave shells and one convex shell atleast thrice.

(A) $\frac{32}{371}$ (B) $\frac{32}{375}$ (C) $\frac{1}{5}$ (D) $\frac{2}{5}$

7. If n is an even natural number and coefficient of x^r in the expansion of $\frac{(1+x)^n}{1-x}$ is 2^n ($|x| < 1$), then which of the following may be true ?

(A) $r \leq \frac{n}{2}$ (B) $r \geq \frac{n-2}{2}$ (C) $r > n$ (D) $r = n$

8. If $ac > b^2$ then the sum of the coefficients in the expansion of $(a\alpha^2x^2 + 2b\alpha x + c)^n$ where $a, b, c, \alpha \in \mathbb{R}$ and $n \in \mathbb{N}$, is

(A) positive if $a > 0$ (B) positive if $c > 0$ (C) negative if $a < 0$ (D) positive if $c < 0$, n is even

9. If $(1-x^3)^n = \sum_{r=0}^n a_r x^r (1-x)^{3n-2r}$, then (where $n \in \mathbb{N}$)

(A) $\sum_{r=0}^n a_r = 4^n$ (B) $\sum_{r=0}^n (-1)^r a_r = 2^n$ (C) $\sum_{r=0}^n r a_r = 3n4^{n-1}$ (D) $\sum_{r=0}^n \frac{a_r}{r+1} = \frac{4^{n+1}-1}{3(n+1)}$

10. Which of the following statements are correct

(A) $\sum_{k=1}^n k^2 ({}^n C_k)^2 = n^2 ({}^{2n-2} C_{n-1})$

(B) ${}^{674} C_0 {}^{2022} C_{674} - {}^{674} C_1 {}^{2020} C_{674} + {}^{674} C_2 {}^{2018} C_{674} - \dots + {}^{674} C_{674} {}^{674} C_{674} = 2^{674}$

(C) $\sum_{i=0}^n \sum_{j=0}^n {}^n C_i \cdot {}^n C_j = 2^{2n}$

(D) $\sum_{i>j \geq 0} {}^{n+1} C_i \cdot {}^n C_j = 2^{2n}$

11. Let the equation $x^{100} + a_{99}x^{99} + a_{98}x^{98} + \dots + a_1x + a_0 = 0$ has roots ${}^{99}C_0, {}^{99}C_1, {}^{99}C_2, \dots, {}^{99}C_{99}$, then

(A) $a_{99} = 2^{99}$

(B) $a_{98} = \frac{2^{198} - {}^{198} C_{99}}{2}$

(C) $({}^{99}C_0)^2 + ({}^{99}C_1)^2 + \dots + ({}^{99}C_{99})^2 = {}^{198} C_{99}$

(D) $({}^{99}C_0)^2 + ({}^{99}C_1)^2 + \dots + ({}^{99}C_{99})^2 = a_{99}^2 - {}^2 a_{98}$

12. Which of the following statements are correct
 (A) Last two digit of the number 19^{94} is 19
 (B) Last four digits of the number $7^{100} - 3^{100}$ is 4000
 (C) The remainder when $(3053)^{456} - (2417)^{333}$ is divisible by 9 is 2
 (D) $19^{93} - 13^{99}$ is divisible by 162
13. Let the set $A = [T(r+1) = {}^nC_r (3)^{n-r} (5x)^r, r = 0, 1, 2, 3, \dots, n]$, then
 (A) For $x = 1$, if $T(12)$ is the greatest, then the value of n can be 17
 (B) For $n = 25$, if $T(21)$ is the greatest, then the value of $5x$ can be 12
 (C) For $n = 20$, if $T(10) = T(11)$, then the value of $33x$ can be 18
 (D) For $x = 1$, if $T(12)$ is the greatest, then the value of n can be 19
14. The number of ways to choose a subset of two elements $\{a, b\}$ from the set $\{1, 2, 3, \dots, 49, 50\}$ such that (consider $\{a, b\}$ and $\{b, a\}$ alike)
 (A) $|a-b| \leq 5$ is 245 (B) $|a-b| \leq 5$ is 235 (C) $|a-b| = 5$ is 45 (D) $|a-b| \geq 5$ is 1035
15. The number of ways we can form a quadrilateral by joining the vertices of 'n' sided regular polygon such that ($n \geq 6$)
 (A) quadrilateral has three sides common with the polygon is n
 (B) quadrilateral has three sides common with the polygon is $n - 3$
 (C) quadrilateral has odd number of sides common with the polygon is $n [{}^{n-4}C_2 - n + 6]$
 (D) quadrilateral has odd number of sides common with the polygon is $n [{}^{n-4}C_2]$
16. The number of seven digit numbers $x_1x_2x_3x_4x_5x_6x_7$ such that
 (A) $x_1 < x_2 < x_3 < x_4 < x_5 < x_6 < x_7$ is 9C_7 (B) $x_1 \leq x_2 < x_3 < x_4 < x_5 < x_6 < x_7$ is ${}^{10}C_7$
 (C) $x_1 \leq x_2 < x_3 < x_4 \leq x_5 < x_6 < x_7$ is ${}^{11}C_7$ (D) $x_1 > x_2 > x_3 > x_4 > x_5 > x_6 > x_7$ is ${}^{10}C_7$
17. If $\sum_{1 \leq a < b < c} \frac{1}{2^a 3^b 5^c} = \frac{1}{n}$, then :
 (A) The number of divisors of 'n' are 16 (B) The sum of divisors of 'n' is 3600
 (C) The number of divisors of 'n' are 12 (D) Number of zeros at the end of $n!$ is 402
18. The number of ways we can distribute 999 balls in 3 boxes. Such that (box can be empty also)
 (A) if boxes & balls are distinct, then number of ways are 3^{999}
 (B) if boxes & balls are distinct, then number of ways are 999^3
 (C) if balls are identical & boxes are different, then number of ways are 500500
 (D) if balls are identical & boxes are identical, then number of ways are 83667
19. Let $P_1, P_2, \dots, P_{15}$ be 15 points on a circle. The number of distinct triangles formed by points P_i, P_j, P_k such that $i + j + k \neq 15$, is
 (A) 443
 (B) 455
 (C) If no two consecutive points are chosen, then the number of triangles is 268
 (D) If exactly two sides are common with polygon made by these points, then the number of triangles is 14

20. There are 'n' locks and n corresponding keys are available but actual combination is not known. A_n is the number of trials that are required to assign the keys such that no lock will open and B_n is the maximum number of trials that are needed to assign the keys to their corresponding locks.

Which of the following is(are) true.

(A) $A_6 + B_6 = 280$ (B) $A_7 + B_7 = 1849$ (C) $B_n = {}^nC_2$ (D) $B_n = \sum_{k=2}^n (k-1)$

21. In a game, a coin is tossed $(2n + m)$ times and a player wins if he does not get any two consecutive outcomes same for atleast $2n$ times in a row. The probability that player wins the game is $\lambda(m, n)$, then

(A) $\lambda(10, 5) = \frac{3}{256}$ (B) $\lambda(6, 6) = \frac{1}{512}$
 (C) $\lambda(10, 6) = \frac{5}{2048}$ (D) $\lambda(14, 7) = \frac{1}{1024}$

22. For the 3 events A, B and C, $P(\text{atleast one occurring}) = \frac{3}{4}$, $P(\text{atleast two occurring}) = \frac{1}{2}$ and $P(\text{exactly two occurring}) = \frac{2}{5}$. Which of the following relations is/are CORRECT?

(A) $P(ABC) = \frac{1}{10}$ (B) $P(AB) + P(BC) + P(CA) = \frac{7}{10}$
 (C) $P(A) + P(B) + P(C) = \frac{27}{20}$ (D) $P(\overline{A}\overline{B}\overline{C}) + P(\overline{A}\overline{B}C) + P(\overline{A}B\overline{C}) = \frac{1}{4}$

23. There are four parcels namely A, B, C and D in a room. A particular I-phone is in one of them. Then probability that it is in parcel A is $\frac{2}{5}$, but if it is not in parcel A its probability is equally likely to be either B or C or D.

I sent any delivery man to bring one of a parcel to me. The probability that my delivery man brings the parcel A is $\frac{2}{3}$ and the probability that he brings any other parcel is $\frac{1}{3}$. Let α = I get the particular I-phone

β = my delivery man brings parcel A

(A) $P(\alpha) = \frac{1}{3}$ (B) $P(\alpha) = \frac{7}{15}$ (C) $P\left(\frac{\beta}{\alpha}\right) = \frac{3}{7}$ (D) $P\left(\frac{\beta}{\alpha}\right) = \frac{4}{7}$

24. One die has three faces marked 1, two faces marked 2 and one face marked 3. Another has one face marked 1, two faces marked 2 and three faces marked 3 then

- (A) The most probable throw with two dice is 4
 (B) The probability of most probable throw with two dice is $\frac{1}{4}$
 (C) The probability of most probable throw with two dice is $\frac{7}{18}$
 (D) The most probable throw with two dice is 5

25. There is a group of 6 persons. They play a game in which each has to select a number from 1 to 4. Let A_n is event that n persons have selection of same number, then

(A) $P(A_5) = \frac{18}{4^6}$ (B) $P(A_5) = \frac{18}{4^5}$ (C) $P(A_6) = \frac{1}{4^5}$ (D) $P\left(\frac{A_5}{A_6}\right) = \frac{1}{2}$

26. Let a function f is defined as $f : \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4\}$. If f satisfy $f(f(x)) = f(x) \forall x \in \{1, 2, 3, 4\}$. Then which of the following is(are) true

- (A) Number of such functions are 41
 (B) Number fo such functions are 39
 (C) Selecting a one-one function is equally likely as selection of a onto function
 (D) probability of selecting a bijective function is $\frac{1}{41}$

27. Let $(1 + ax + bx^2 + cx^3)^{10} = 1 + P_1x + P_2x^2 + P_3x^3 + \dots + P_{30}x^{30}$. If $P_1 = 20$, $P_2 = 200$, $P_3 = 1000$, then

Column-I

Column-II

- | | |
|--|--------|
| (A) $P_1 + P_2 + P_3 + \dots + P_{30} = 27^\alpha - 1$, where α is | (p) 4 |
| (B) $\frac{P_4}{3(a+b)}$ | (q) 8 |
| (C) $\frac{ c }{a+b}$ | (r) 10 |
| (D) $a+b$ | (s) 55 |

28. Let M is a 3×3 matrix whose entries are taken from the set $\{0, 1, 2, 3, 4\}$ such that $\text{tr}(M) = 5$, then (where $\text{tr}(M)$ denotes trace of matrix M)

Column-I

Column-II

- | | |
|--|---------------|
| (A) Number of such matrices 'M' are | (p) 18 |
| (B) If M is diagonal matrix then number of such matrices are | (q) $18(5^6)$ |
| (C) If M is symmetric matrix then number of such matrices are | (r) 2250 |
| (D) If M is lower triangular matrix then number of such matrices are | (s) 5^6 |

29. If $(1 + x + x^2)^{2023} = a_0 + a_1x + a_2x^2 + \dots + a_{4046}x^{4046}$ and $a_0 + a_3 + a_6 + \dots = n$, then the number of divisors of 'n' are

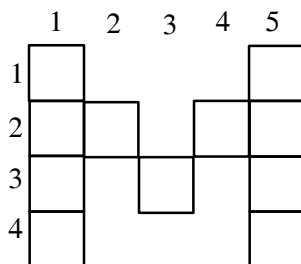
30. Find greatest possible value of n if ${}^9C_n + 3({}^9C_{n+1}) + 3({}^9C_{n+2}) + {}^9C_{n+3} > {}^{12}C_{n+2}$

31. The coefficient of x^8 in the expansion of $(2 + 9x^3 + 6x^4 + x^5)^{10}$ is $5 \cdot 2^p \cdot 3^q$, where $p, q \in \mathbb{N}$, then $\frac{p+q}{5}$ is

32. If $(1+x)^2 \left(1+\frac{x}{2}\right)^2 \left(1+\frac{x}{2^2}\right)^2 \left(1+\frac{x}{2^3}\right)^2 \dots = P_0 + P_1x + P_2x^2 + \dots$ then find the value of $3P_2$

33. Mr. A can climb either one step or two steps at a time. Let a_n is the number of ways he climbs a n-step stair case, then a_{11} is

34. Find the number of permutations of the digits 1,2,3,4,5 taken all at a time so that the sum of the digits at the first two places is smaller than the sum of the digits at the last two places.
35. In how many ways 6 identical black balls and 6 identical white balls be arranged in a row such that on picking any 3 consecutive balls from the arrangement, we get at least one black ball.
36. Number of ways in which first 11 natural numbers can be arranged in the given figure such that $a_{ij} > a_{kj}$ whenever $i > k \forall j$ is (a_{ij} represents elements in i^{th} row and j^{th} column)



37. On each evening a boy either watches Discovery channel or MNX. The probability that he watches MNX is $\frac{4}{5}$. If he watches Discovery, there is a probability of $\frac{3}{4}$ that he will fall asleep, while it is $\frac{1}{4}$ when he watches MNX. On one evening, the boy is found asleep while watching TV. The probability that he watched Discovery is $\frac{m}{n}$ where m, n are co-prime natural numbers. Find $|m-n|$
38. There are five students standing in a line with their hats. Suddenly the wind picks up the hats and randomly assigns each hat to a student, the probability that no students will get his/her own hat is P , then $3P$ is equal to
39. Many states use a sequence of three letters followed by a sequence of three digits as then standard license-plate pattern. Given that each three-letter three-digit arrangement is equally likely, the probability that such a license plate will contain at least one palindrome (a three-letter arrangement or a three-digit arrangement that reads the same left-to-right as it does right-to-left) is $\frac{m}{n}$, where m and n are relatively prime positive integers, then the value of $m + n$
40. In a hurdle race, a runner has probability p of jumping over a specific hurdle. Given that in 5 trials, the runner succeeded 3 times, the conditional probability that the runner has succeeded in the first trial is _____.
41. Let p, q be chosen one by one from the set $\{1, \sqrt{2}, \sqrt{3}, 2, e, \pi\}$ with replacement. Now a circle is drawn by taking (p, q) as its centre. Then the probability that at most two rational points (both coordinates rational) exist on the circle $\frac{p}{q}$, where p and q are coprime, then $p + q$ is equal to

42. A slip of paper is given to a person 'A' who marks it with either a plus or a minus sign. The probability of his writing plus is $\frac{1}{3}$. 'A' passes the slip to 'B' who may either leave it as it is or change the sign before passing it to 'C'. Similarly 'C' passes it on to 'D' and 'D' passes it to a referee who sees the plus sign on the slip. If 'B', 'C', 'D' change the sign with probability $\frac{2}{3}$, then the probability that A originally wrote the plus sign is P, then the value of 82P is
43. The contents of 3 bags w.r.t green and red marble is as given in the table shown.

Bag	G	R
A	3	1
B	2	2
C	1	3

A child randomly selects one of the bags, and draws a marble from it and retains it. If the marble is green, the child draws the second marbles randomly from one of the two remaining bags. If the first marble drawn is red the child draws one more marble from the same bag. The probability that the second drawn marble is green is expressed as $\frac{m}{n}$ (where m and n are coprime). Find the value of $(2m - n)$.

ANSWER KEY

1. (A) 2. (B) 3. (D) 4. (A) 5. (D) 6. (A) 7. (CD)
8. (ABD) 9. (ACD) 10. (ABCD) 11. (BCD) 12. (ACD) 13. (ABC)
14. (BCD) 15. (AC) 16. (ABCD) 17. (ABD) 18. (ACD) 19. (ACD)
20. (ACD) 21. (ABD) 22. (ABCD) 23. (BD) 24. (AC) 25. (BC)
26. (ACD) 27. $(A \rightarrow (r), (B \rightarrow (s), (C \rightarrow (q), (D \rightarrow (p)$
28. $(A \rightarrow (q), (B \rightarrow (p), (C \rightarrow (r), (D \rightarrow (q)$ 29. (2023) 30. (3) 31. (2.80)
32. (20) 33. (144) 34. (48) 35. (357) 36. (69300) 37. (4)
38. (1.1) 39. (59) 40. (0.60) 41. (17) 42. (26) 43. (2)